TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Modular Energy Management and Calibration Toolkit for Electrified Powertrains with Battery State Estimation			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Bio/Health ( ), Hydrogen ( ), Future Mobility ( ), Cyber Security ( ), AI ( ), Next Generation Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical & Electronics		Information & Communications	
	Medium Category	Small Category	Medium Category	Small Category
	Batteries	Secondary Batteries	Software	Embedded SW
Concept of Proposed Technology	Automotive control and energy management systems are **central to modern vehicle performance, efficiency, and sustainability**. These systems coordinate how energy is distributed and utilized across powertrain components—such as batteries, electric motors, and engines—ensuring that vehicles meet performance demands while minimizing energy losses and emissions. Effective energy management directly influences **fuel economy, battery usage, and range** in hybrid and electric vehicles by optimizing the flow and use of available energy in real time. In electrified vehicles, advanced control strategies are essential not only for balancing power delivery between subsystems but also for meeting regulatory efficiency targets and			

customer expectations for range and durability.

TNO's Modular Energy Management Strategy (MEMS) is designed to address a set of recurring engineering and R&D bottlenecks that show up when you move from a single powertrain configuration to many variants, and when you want something closer to optimal control than hand-tuned rules.

1) Exploding complexity from many architectures and component variants

Challenge: Sustainable powertrains come in many configurations (battery electric, fuel cell, hybrid, different component choices), and each application often needs a different configuration for best total cost of ownership and productivity. This creates a combinatorial burden in architecture selection and controller development.

How MEMS helps: MEMS uses a modular structure so engineers can swap components and topologies while keeping a consistent controller structure, making it practical to compare many configurations quickly and consistently.

2) High development time and cost for energy management calibration per topology

Challenge: Developing an energy management system for each vehicle and application is described by TNO as time-consuming and costly because of the multitude of drive and energy components and variants.

How MEMS helps: TNO positions MEMS as a method and software suite that keeps development complexity manageable and reduces time-to-market. They also report quantified time savings per topology and additional savings from reuse because components do not need to be recalibrated each time.

3) Lack of reuse and scalability in traditional heuristic or topology-specific controllers

Challenge: Many heuristic or rule-based strategies are tailored to one specific problem or topology and are not reusable across different architectures.

How MEMS helps: In the EVS35 paper, MEMS is described as applying dual decomposition to an ECMS optimal control formulation, splitting the problem into smaller subsystem problems. This yields a scalable and reusable approach across configurations when the power balance structure is known.

4) Difficulty of doing "co-design" (optimize hardware plus control) at scale

Challenge: Electrification opens a big design space in topology, sizing, and control; automatically producing optimized control strategies is valuable because it reduces development time and cost, but it is hard to do consistently across many candidates.

How MEMS helps: TNO explicitly positions MEMS as supporting both (a) exploring the design space and (b) automatically producing optimized control algorithms within a co-design workflow.

5) Managing battery stress and peak power demands while meeting mission needs

Challenge: High-demand events can drive battery stress (high peak powers), which affects durability and thermal constraints in real applications.

How MEMS helps: In TNO's co-design publication entry, results are reported where the MEMS controller reduces battery stress by peak-shaving battery power during high-demand periods.

6) Practical deployability on real vehicle control units

Challenge: Even if an algorithm is good in simulation, it still has to be implementable on a VCU in real time.

How MEMS helps: TNO states that MEMS can efficiently run on vehicle control units and that it has evolved into a broader software suite and methodology that partners can learn to use independently.

Proposed core technology:

• Modular Energy Management Strategy (MEMS): an optimization-based supervisory control framework. In published work, MEMS is described as using dual decomposition to solve the Equivalent Consumption Minimization Strategy (ECMS) optimal control problem in a modular way, enabling scalable integration of multiple subsystems and topologies and efficient real-time execution. [Ref.2-3]

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Rationale for Joint R&D Project

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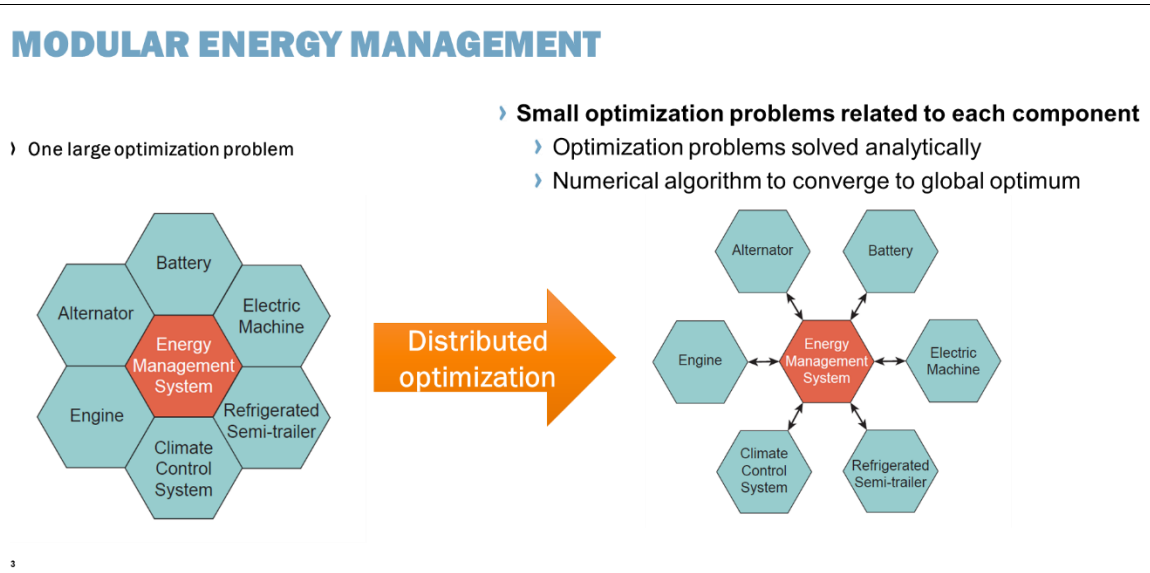

• Predictive energy management extensions: leverage route preview information (for example speed limits, road slope, traffic) to optimize energy use. Published results report potential energy savings of 7.1% in simulation for a multi-layer predictive energy management system for BEVs. [Ref.6]

• Battery modelling and BMS algorithm toolchain: SoX estimation (SoC, SoH, thermal state), in-situ parameter identification, balancing algorithm development, ageing models, and physics-model-based control and optimization. [Ref.4,9]

• Calibration workflow integration: a reusable workflow linking Model-in-the-Loop (MIL), Hardware-in-the-Loop (HIL) and vehicle testing to accelerate calibration and reduce risk before on-road deployment. [Ref.8-9]

Planned implementation form:

• A Simulink-based reference implementation and libraries suitable for rapid prototyping, with defined interfaces for deployment to embedded controllers used in vehicle energy management and calibration projects.

Why joint R&D is needed:

• The Korean SME partner is a vehicle engineering and powertrain calibration specialist seeking differentiation in electrified powertrain development where supervisory control, battery state estimation and efficient calibration are key.

• TNO contributes proprietary MEMS know-how, battery modelling and BMS algorithm development expertise, and validation facilities and methods.

• The SME contributes industrial integration and calibration workflows, and access to Korea-relevant use cases and demonstrators.

• Collaboration is required to adapt the toolkit to Korea-relevant missions (traffic, topology, climate and charging behaviour), integrate it into the SME toolchain, and validate performance and robustness on representative platforms. [Ref.1-3,5-9]

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Key Words

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1. Modular Energy Management Strategy (MEMS)

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	2. Predictive energy management 3. Battery SoX state estimation 4. Degradation-aware control 5. Model-based calibration (MIL/HIL)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (7)
R&D Objectives	R&D Objectives: • Adapt MEMS to the partner's target electrified powertrain topologies and constraints (driveability, energy, battery limits and mission constraints). [Ref.2-3] • Develop and calibrate battery state estimators (SoC, SoH and thermal state) suitable for real-time deployment, including parameter identification and thermal interfaces. [Ref.4,9] • Develop degradation-aware supervisory control terms and charge/mission planning hooks that account for battery ageing and operational cost. [Ref.5,7] • Develop a model-based calibration workflow integrating MIL and HIL with vehicle data to reduce calibration effort and improve transferability across platforms. [Ref.8-9] • Demonstrate the integrated toolkit on representative demonstrators and validate improvements against a baseline.			
R&D Target Specifications	R&D Target Specifications (targets to be finalized in WP1): • Energy consumption reduction: target 5% to 7% versus a non-predictive baseline in representative missions (informed by published predictive energy management results reporting 7.1% energy savings in simulation). [Ref.6] • Battery stress reduction: target 10% reduction in peak battery power and/or equivalent C-rate excursions during high-demand events, while meeting performance constraints. • Real-time deployability: supervisory control cycle time compatible with embedded vehicle control units via modular decomposition. [Ref.2-3] • State estimation: SoC estimation target <2% RMS after calibration on representative duty cycles; thermal state estimation integrated with pack thermal constraints. [Ref.4,9]			
R&D Plan	R&D Plan (indicative 36 months): WP1 (M0-M6): System definition and baseline: define target powertrain architectures, constraints, missions and KPIs; collect required data; establish baseline controller and models. WP2 (M4-M14): Battery modelling and state estimation: model selection, parameter identification, SoX estimators, thermal model integration; validation in MIL and initial HIL bring-up. WP3 (M10-M22): MEMS adaptation and control integration: modularization for target topologies; constraint handling; degradation-aware terms; integration with charging/mission planning. WP4 (M18-M30): HIL and lab validation: implement controller and estimators in HIL; execute robustness and stress tests; iterate calibration and model updates. WP5 (M26-M36): Demonstrator and exploitation: deploy on a representative platform; validate			

	KPIs; finalize documentation and exploitation plan.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Research Institute (TNO) expected R&R: • Provide MEMS-based supervisory control concepts and algorithm prototypes (simulation-ready and real-time-ready variants). [Ref.2-3] • Develop battery models and SoX estimators (including parameter identification, balancing-related modelling and thermal prediction/control interfaces). [Ref.4,9] • Provide MIL/HIL validation methodology and execute joint validation activities using TNO facilities where applicable. [Ref.8-9] • Lead scientific documentation and (where agreed) joint publications and IP generation.
Expected R&R of Korean SME Partner	Korean SME Partner expected R&R: • Define use cases, target platforms and calibration constraints; provide relevant models and test data (subject to NDA and data governance). • Integrate the toolkit into the SME engineering toolchain and support embedded deployment on target hardware. • Execute Korea-based validation on partner demonstrators and provide feedback for iterative refinement. • Lead exploitation planning for commercial engineering services and potential licensing or joint offerings.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Expected Outcomes: • Validated modular energy management and calibration toolkit for electrified powertrains, with documented interfaces and reusable model and controller components. • Demonstrated improvements in mission energy efficiency and reduced battery stress, supporting improved range and battery lifetime potential (to be quantified in the project). • Reduced calibration effort through a reusable MIL/HIL-to-vehicle workflow, improving time-to-market for new electrified platforms. • Optional module for cost-aware charge or mission planning (for example fleet charging schedules) aligned with published approaches. [Ref.7] • Jointly exploitable know-how and potential IP, enabling commercialization through the Korean SME partner's engineering services. Expected Impact: This proposal supports advanced mobility and secondary battery competitiveness by improving vehicle energy efficiency, robustness of battery management, and industrial deployment speed of optimization-based control. [Ref.1-9]

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization